

Generation Interconnection Request System Impact Study Report

For Queue Project Q11-0025

Walerga- Antelope 138 kV
March 25, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0025. The Interconnection Customer proposes a 826.3 MW natural gas combined cycle facility interconnecting at the Walerga- Antelope 138 kV substation in the Virginia area. The request was received on April 17, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 21 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$449,498,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The North Huntington to Walerga- Antelope 138kV is overloaded to 344.7% of its 236 MVA emergency rating for the loss of the Moorland to Walerga- Antelope 138kV. The project contributes 100.0% DFAX (826.3 MW) to this overload.

For the loss of Walerga- Antelope to North Huntington 138kV L, the Walerga- Antelope to Moorland 138kV L loads to 336.2% of its 242 MVA rating. Project DFAX: 100.0%.

For the loss of Walerga- Antelope to N Huntington 138kV, the Penhook to Island Falls 138kV loads to 283.4% of its 190 MVA rating. Project DFAX: 57.2%.

For the loss of North Huntington to Walerga- Antelope 138kV L, the Island Falls to Moorland 138kV loads to 278.7% of its 195 MVA rating. Project DFAX: 57.2%.

For the loss of North Huntington to Florida 138kV, the N Huntington to Enslen 138kV Line loads to 223.4% of its 201 MVA rating. Project DFAX: 43.8%.

The Leesville Substation to Penhook 138kV is overloaded to 145.9% of its 221 MVA emergency rating for the loss of the N Huntington to Walerga- Antelope 138kV. The project contributes 40.4% DFAX (334.1 MW) to this overload.

The S Colton to Otter 138kV Line 2 is overloaded to 144.5% of its 284 MVA emergency rating for the loss of the S Colton to Otter 138kV. The project contributes 16.3% DFAX (134.6 MW) to this overload.

Otter to S Colton 138kV Line 1: 141.9% loading (289 MVA) following the outage of the Otter to South Colton 138kV #2; DFAX = 16.3%.

For the loss of Reusens to Huntington Court 345kV, the Skiffes Crk to Michigan Power Lp 138kV L loads to 127.6% of its 271 MVA rating. Project DFAX: 15.8%.

The Enslen to Otter 138kV #1 is overloaded to 126.7% of its 304 MVA emergency rating for the loss of the Enslen to Otter 138kV Cir. 2. The project contributes 15.0% DFAX (124.3 MW) to this overload.

S Crk Substation to Aibonito Pds 138kV: 124.5% loading (260 MVA) following the outage of the Clover to Bear 345kV; DFAX = 7.7%.

For the loss of Enslen to Otter 138kV Ckt 1, the Ensln to Otter 138kV Circuit 2 loads to 123.3% of its 313 MVA rating. Project DFAX: 15.0%.

For the loss of Huntington Court to Reusens 345kV Line, the North Huntington Substation to Florida 138kV loads to 122.0% of its 305 MVA rating. Project DFAX: 15.8%.

For the loss of N Huntington to Walerga- Antelope 138kV, the Otter River to Belagio Substation 138kV loads to 120.7% of its 255 MVA rating. Project DFAX: 18.6%.

For the loss of Enslens to Cristianitos 138kV line, the Yaphank to South Colton 138kV loads to 120.6% of its 289 MVA rating. Project DFAX: 9.3%.

The Flowood to Otter Riv 138kV is overloaded to 119.2% of its 251 MVA emergency rating for the loss of the North Huntington to Walerga- Antelope 138kV. The project contributes 18.6% DFAX (153.8 MW) to this overload.

The Brush Tavern to Rustburg 138kV L is overloaded to 118.4% of its 243 MVA emergency rating for the loss of the Cover to Bear 345kV. The project contributes 11.0% DFAX (91.0 MW) to this overload.

Berlin Gorham to Skiffes Creek 138kV: 117.1% loading (306 MVA) following the outage of the Huntington Court to Reusens Substation 345kV; DFAX = 15.8%.

For the loss of Bear to Clover 345kV, the Monel to Rustburg 138kV loads to 113.9% of its 223 MVA rating. Project DFAX: 11.0%.

Bear to Leavitt 345kV Circuit 2: 112.7% loading (1,510 MVA) following the outage of the Leavitt to Bear 345kV Circuit 1; DFAX = 16.6%.

Desert Outpost to S Texa 138kV line: 112.4% loading (247 MVA) following the outage of the Brush Tavern to Rustburg 138kV Line; DFAX = 8.9%.

Russell Fork to Robison Park 230kV: 112.3% loading (557 MVA) following the outage of the Jones Chapel Substation to Dekalb 500kV; DFAX = 7.8%.

Florida to Berlin Gorham 138kV: 109.9% loading (334 MVA) following the outage of the Reusens to Huntington Court 345kV; DFAX = 15.8%.

The Leavitt to Birchwood 345kV is overloaded to 109.8% of its 1,384 MVA emergency rating for the loss of the Fluvanna P.S to Louisa Generation Facility 345kV. The project contributes 16.9% DFAX (139.4 MW) to this overload.

The Michigan Power Lp to Blue Ridge 138kV is overloaded to 109.8% of its 255 MVA emergency rating for the loss of the Reusens to Huntington Court 345kV line. The project contributes 6.6% DFAX (54.2 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
North Huntington to Walerga- Antelope 138kV	Rebuild line with higher-capacity conductor	\$3,347,000	100% (\$3,347,000)
Walerga- Antelope to Mooland 138kV Line	Rebuild line with higher-capacity conductor	\$14,471,000	100% (\$14,471,000)
Penhook to Island Fals 138kV	Rebuild line with higher-capacity conductor	\$5,878,000	100% (\$5,878,000)
Moorland Substation to Island alls 138kV	Rebuild line with higher-capacity conductor	\$8,672,000	100% (\$8,672,000)
N Huntington to Enslens 138kV Line	Rebuild line with higher-capacity conductor	\$19,744,000	100% (\$19,744,000)
Penhook to Leesville 138kV	Rebuild line with higher-capacity conductor	\$15,282,000	100% (\$15,282,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
S Colton to Otter 138kV Circuit 2	Rebuild line with higher-capacity conductor	\$8,333,000	100% (\$8,333,000)
S Colton to Otter Substation 138kV Line 1	Rebuild line with higher-capacity conductor	\$7,072,000	100% (\$7,072,000)
Skiffes Crk to Michigan Power Lp 138kV L	Rebuild line with higher-capacity conductor	\$5,424,000	100% (\$5,424,000)
Otter to Enslen 138kV	Rebuild line with higher-capacity conductor	\$7,203,000	100% (\$7,203,000)
S Crk Substation to Aibonito Pds 138kV	Rebuild line with higher-capacity conductor	\$7,738,000	100% (\$7,738,000)
Enslen to Otter 138kV Circuit 2	Rebuild line with higher-capacity conductor	\$11,700,000	100% (\$11,700,000)
N Huntington to Florida Substation 138kV L	Rebuild line with higher-capacity conductor	\$9,323,000	100% (\$9,323,000)
Otter Riv to Belagio 138kV	Rebuild line with higher-capacity conductor	\$3,399,000	100% (\$3,399,000)
S Colton to Yaphank 138kV L	Rebuild line with higher-capacity conductor	\$7,586,000	100% (\$7,586,000)
Flowood to Otter Riv 138kV	Rebuild line with higher-capacity conductor	\$10,332,000	100% (\$10,332,000)
Brush Tavern to Rusturg 138kV	Rebuild line with higher-capacity conductor	\$10,917,000	100% (\$10,917,000)
Berlin Gorham to Skiffes Creek 138kV	Rebuild line with higher-capacity conductor	\$11,572,000	100% (\$11,572,000)
Monel to Rustburg 138kV	Rebuild line with higher-capacity conductor	\$16,001,000	100% (\$16,001,000)
Bear to Leavitt 345kV L	Rebuild line with higher-capacity conductor	\$165,450,000	100% (\$165,450,000)
Desert Outpost to S Texa 138kV line	Rebuild line with higher-capacity conductor	\$1,268,000	100% (\$1,268,000)
Robison Park to Russell Fork 230kV Line	Rebuild line with higher-capacity conductor	\$43,934,000	100% (\$43,934,000)
Florida to Berlin Gorham 138kV	Reconductor line	\$1,240,000	100% (\$1,240,000)
Leavitt to Birchwood 345kV	Reconductor line	\$51,397,000	100% (\$51,397,000)
Michigan Power Lp to Blue Ridge 138kV	Reconductor line	\$2,215,000	100% (\$2,215,000)
	Total allocated to project		\$449,498,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.